Viscosity of Fluids

Group 4
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Chemical Engineering Lab I
17 October 2019

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Introduction

- The main purpose of this lab is to study the effect of fluid viscosity on flowrate
- Flowrate controlled by pressure to vessel
- Effect of capillary diameter and length studied
- This lab also served to analyze a non-Newtonian fluids behavior

Theory

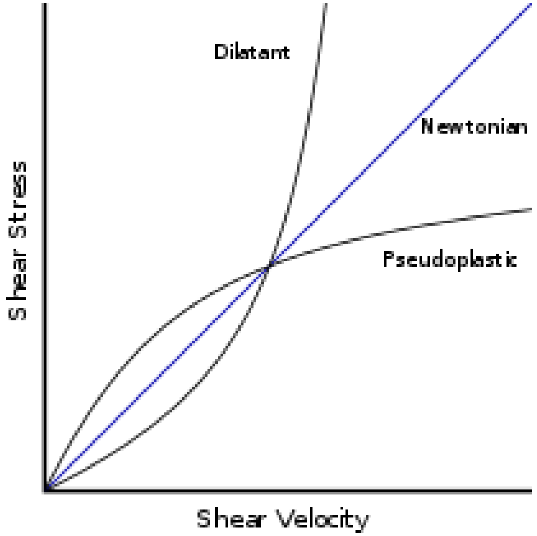
- Newtonian Fluids – linear relationship with viscosity

$$\tau_{yx} = -\mu \frac{dv_x}{dy}$$

- Non-Newtonian Fluids – nonlinear relationship with viscosity

$$\tau_{yx} = -\mu_{app} \frac{dv_x}{dy} = -m \left| \frac{dv_x}{dy} \right|^{n-1} \frac{dv_x}{dy}$$

  - Dilatant – apparent viscosity increases as shear rate increases
  - Pseudoplastic – apparent viscosity decreases as shear rate increases



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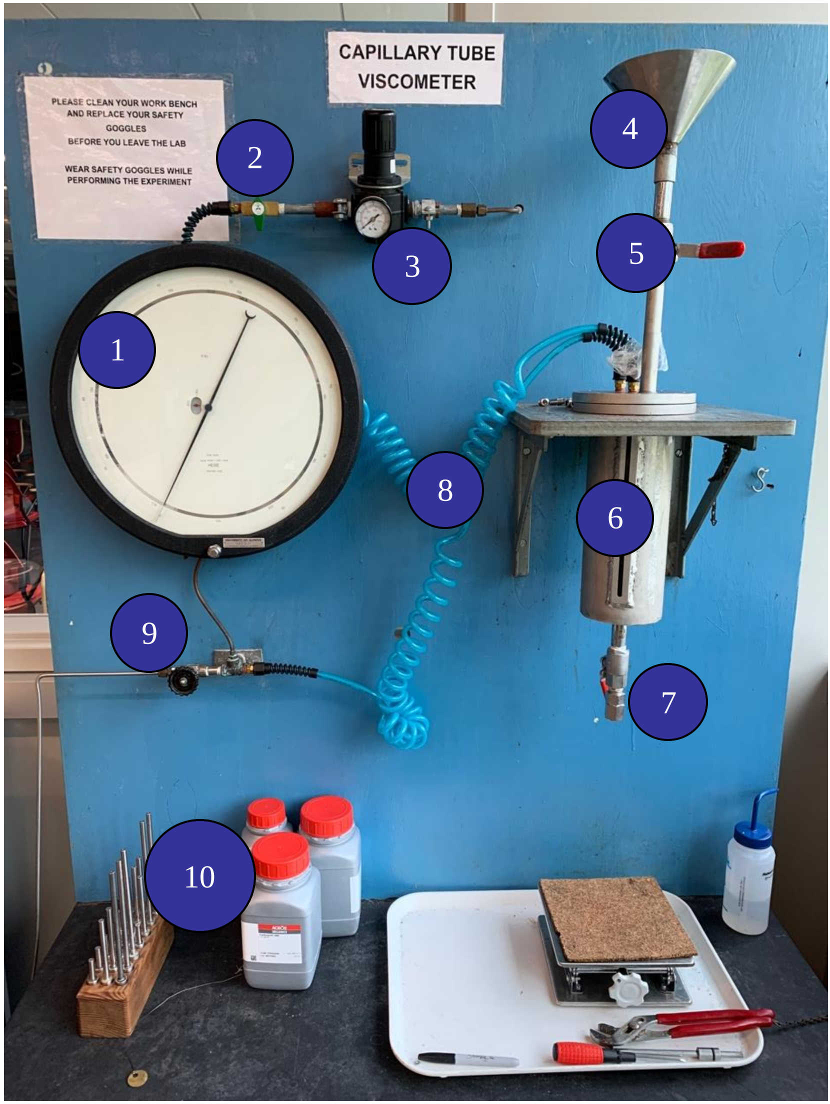
Apparatus

- Main Equipment
  1. Pressure gauge
  2. Valve
  3. Pressure regulator
  4. Loading funnel
  5. Ball valve
  6. Vessel
  7. Ball valve
  8. Hose
  9. Pressure relief valve
  10. Capillary tubes



Material & Supplies

- Material list
  - Warm tap water
  - Carbopol
  - NaOH
  - Scale
  - Mixer
  - Metal container
  - Mechanical lift plate

Procedure

- Prelab Steps
  1. Prepare 0.8 wt% carbopol solution
  2. Measure density of solution
- Lab Day Steps
  1. Load carbopol solution into vessel
  2. Attach capillary tube to vessel
  3. Pressurize vessel and measure mass flowrate through capillary tube
  4. Repeat for 3 total pressures (30 – 40 – 50 psi)
  5. Repeat for 6 total capillary tubes with different lengths and diameters

Experimental Challenges

- Difficulties
  - Mixing carbopol solution properly to eliminate clumps
- Things to watch out for during the experiment
  - Mixer cannot mix solution once NaOH is added – hand mix
  - Start pressure at 30 psi and increase from there
  - Need density to calculate volumetric flowrate
  - Make sure capillary tubes are clean
- Lessons Learned
  - Density of solution shouldn't be less than 1 – check for air bubbles
  - Cleaning vessel can be done without capillary tube at about 5 psi

Safety

- Wear gloves and goggles at all times
- Watch hands and/or any clothes are jewelry with mixer
- Max pressure of system is 60 psi – keep regulator below that

Comment Summary
Page 1
1. 95/100

Page 2
2. Flow rate is the independent variable
3. 13/15

Page 3
4. Show the Ostwald deWaele model
5. 12/15